

0,0 - OVERVIEW

1,0 - FUNDAMENTALS

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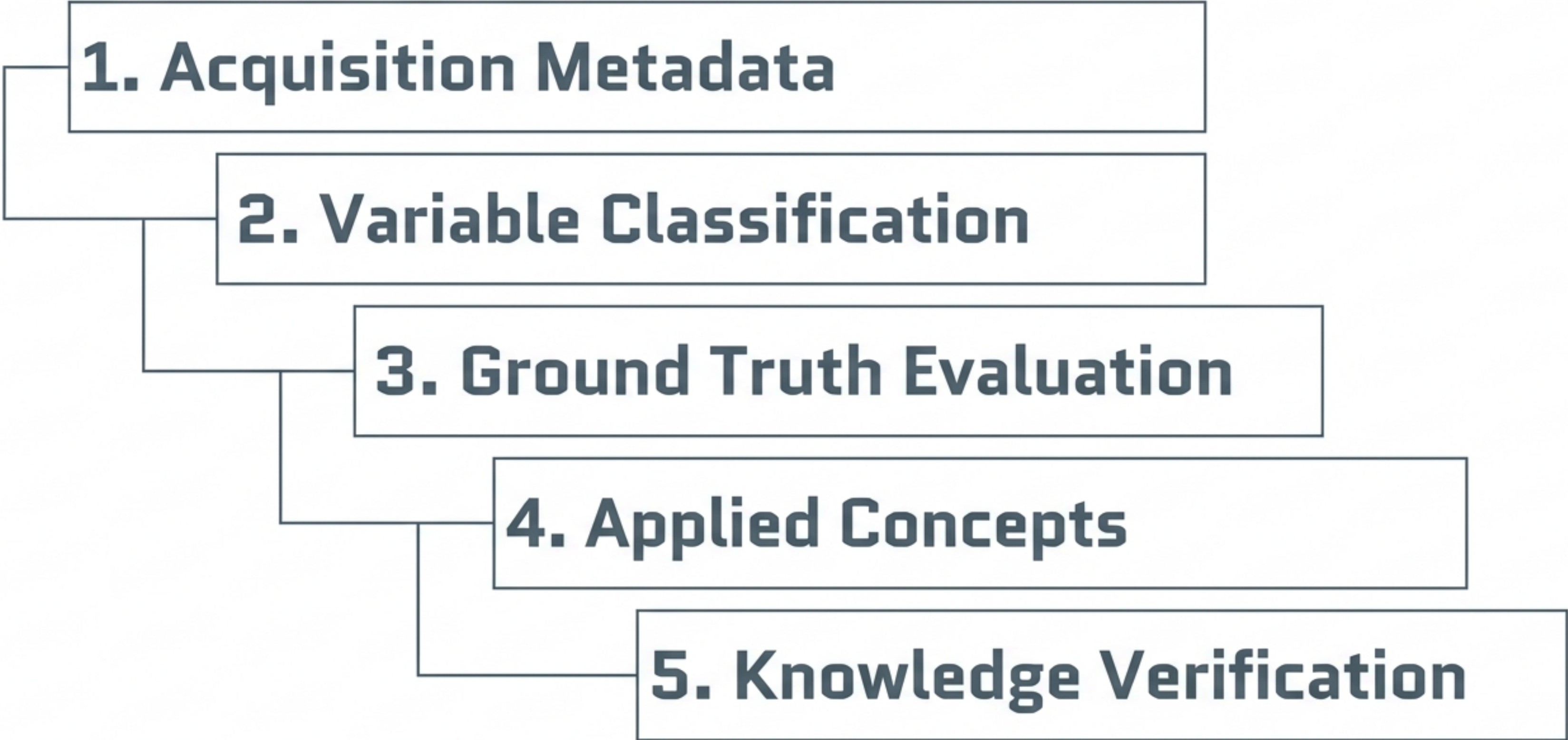
MODULE 3, LESSON 2

PROJECT NO: MENG-AI-003

AI 101 for Mechanical Engineers

Measurements Need Context

Lesson Structure



1. Acquisition Metadata

2. Variable Classification

3. Ground Truth Evaluation

4. Applied Concepts

5. Knowledge Verification

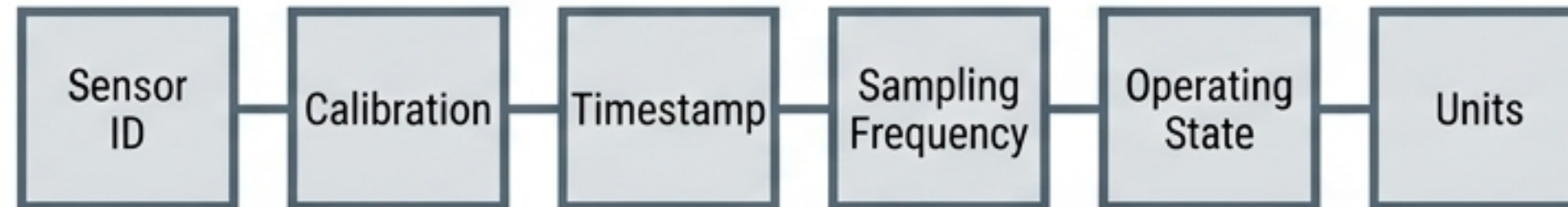
Acquisition Metadata

Defining the context of numerical data.

Signal Context and Acquisition



Acquisition Metadata



Numbers lacking defined units and metadata lack physical interpretation.

Metadata specify exact conditions under which data were obtained.

Raw time samples differ from features computed over defined windows.

Example: Vibration Root Mean Square (RMS) computed over one second.

Variable Classification

Distinguishing physical magnitudes from discrete groups.

Variable Dimensionality Matrices

Numerical Variables (Magnitudes)

Continuous (e.g., Temperature in °C)

Counts (e.g., Mechanical cycles)

Numerical variables quantify physical scales or discrete operational counts.

Categorical Variables (Groups)

Physical State (e.g., Material grade)

System Identifier (e.g., Machine ID)

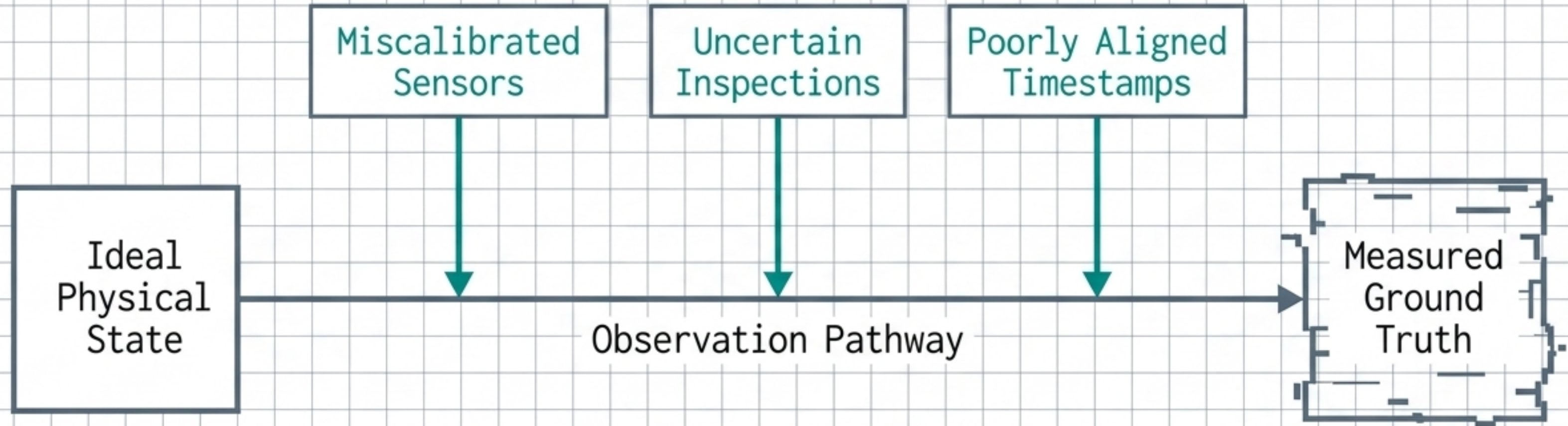
Categorical variables define distinct groups lacking continuous magnitude.

Mathematical distance between categorical IDs carries no physical meaning.

Ground Truth Evaluation

Assessing reference measurement reliability.

Reference Uncertainty Injection



Reference measurements and external annotations define ground truth.

Ground truth data are not inherently error-free facts.

Error vectors inevitably corrupt targeted reference labels.

Engineers must record provenance and verify supported assertions.

Applied Concepts

Standardizing terminology and evaluating physical meaning.

Standard Diagnostic Terminology

Term	Formal Definition	Engineering Example
Ground Truth	Reference value used for comparative analysis	Calibrated but uncertain power measurement
Metadata	Information describing acquisition parameters	Units, calibration dates, sampling frequencies
Categorical Data	Values defining discrete functional groupings	Specific material grades or machine types

Categorical Arithmetic Fallacy



Machine IDs strictly function as categorical group identifiers.

Arithmetic differences between categorical identifiers lack physical interpretation.

Numerical ratios require physical magnitudes to maintain logical validity.

Numbering must explicitly encode defined quantities to permit arithmetic.

Parameter Impact Matrix

Parameter	Meaning	Operational Consequence
Unit	Physical scale	Distinguishes outputs (e.g., W vs kW)
Calibration	Measurement context	Determines baseline reference uncertainty
Machine ID	Category identifier	Invalidates arbitrary numeric ratios
Sampling Rate	Samples per unit time	Defines the relevant signal window

[1] Acquisition Metadata | [2] Variable Classification | [3] Ground Truth Evaluation | [4] Applied Concepts | [5] Knowledge Verification

Knowledge Verification

Validating conceptual acquisition.

Core Axioms Synthesized

1 Metadata contextualize acquisitions but fundamentally do not guarantee causation.

2 Ground truth references intrinsically contain measurement or annotation errors.

3 Categorical identifiers designate classifications devoid of quantifiable physical magnitude.

Key Points

1

Acquisition metadata defines the explicit parameters of numerical measurement environments.

2

System variables mandate strict classification between continuous magnitudes and discrete categories.

3

Ground truth reference measurements inherently possess quantifiable uncertainty and operational error.

4

Categorical variables invalidate continuous mathematical logic and magnitude-based analytical assumptions.

5

Verified knowledge confirms that engineering data requires rigorous contextual qualification.